Motion-Guided Instance OPD

Finite 16-source diagnostic | 14 October 2026

Local LK preserves partial evidence but does not resolve complete tracklet support. Spatial gains remain domain-dependent. Temporal QA and all source checkpoints stay fixed.

Complete-tube m_vIoU (%)

Method	Old HC2	Old Vid	Add HC2	Add Vid
Frozen	22.210	14.490	14.377	3.194
Projection	27.851	22.016	17.384	2.116
Temporal QA (reused)	31.286	22.442	N/A	N/A
Spatial Direct	19.763	16.112	14.377	3.790
Spatial IoU-OPD	19.546	16.447	14.377	3.194
Spatial DIoU-OPD	19.414	16.332	14.377	3.518
QA + Direct	28.539	23.907	N/A	N/A
QA + IoU-OPD	28.306	23.839	N/A	N/A
QA + DIoU-OPD	28.116	23.684	N/A	N/A

Four sources per cell. The additional eight come from the historical pool, not an unseen confirmation cohort. Their Temporal QA outputs do not exist; N/A is not zero. QA + spatial combines a locked old temporal interval with the new actual spatial decode.

Old QA + IoU versus QA: HC2 -2.980 pp; VidSTG +1.397 pp. Additional-eight IoU gives zero average gain in both domains. All negative, neutral and unknown observations are retained.

Execution: 128 teacher calls (96 effective controls), 21 actual fits, 105 declared rounds, 90 actual optimizer steps and 21 final full-model decodes. Ten Source selections reuse F exactly; no new temporal or GT-Oracle fits.

Candidate support and identity controls

Candidate support (includes Source)	Old 8	Additional 8
Total candidates	43	40
At least 2 observation positions	18	17
All 4 positions	11	8
Anchor only	25	23
Missing / all observation slots	88 / 172	86 / 160

Across 67 non-Source candidates: 3 have all four positions; 19 have at least two; 48 retain only the anchor. The old constant-shift construction had 18 complete candidates among the same 43 seeds, versus 11 here. Partial retention is not improved complete-object coverage.

Old seeds reuse the original fixed union; new seeds use a fixed farthest-center grid. Candidate width and height still follow Source. Local flow is a proposal mechanism, not an identity guarantee. Missing geometry and unknown GT are separate. Present-position means and a known-GT common-denominator coverage diagnostic are both published.

Effective factorial control	Same geometry / 16
ID permutation	12
Color permutation	11
Order permutation (fixed badge positions)	14
Joint permutation	11
Wrong query: abstain / different nonzero	14 / 2

All 128 actual responses are valid and untruncated. Thirty-two original order/joint calls also changed badge position; they are retained as superseded controls and replaced by 32 fixed-layout calls. Base selections and locked student references are unchanged.

The six non-Source selections have 4, 2, 1, 1, 1 and 1 valid observation positions. One-anchor selection is not a complete tracklet. Control consistency does not establish semantic identity correctness.

V03 reference isolation and verification

Fixed old reference	vIoU (%)	Delta vs F (pp)
F	6.006629	+0.000000
Source_iou	5.761919	-0.244710
oldR1ID2_iou	8.246636	+2.240006
oldAlternativeID3_iou	1.360450	-4.646179

Same initialization, optimizer, key and five actual steps. Old R1 ID2 repeats its local positive effect; Source and the prelocked old ID3 decrease. This distinguishes these specific references, not every possible perturbation. The current motion-based method chooses Source for V03 and does not update it.

Only one of four reference anchors has GT: Source / old ID2 / old ID3 IoU = 0.445717 / 0.158103 / 0.053349. The other three remain unknown. Sparse reference quality and final complete-tube quality must be evaluated separately.

Independent readbacks

Fresh decode: 64 RGB observation frames and 622 grayscale tracking frames; all 128 rendered inputs and 67 LK trajectories replayed. LK replay uses the same pinned helper. CPU fit math verifies reward weights, likelihood, Adam, source restoration and cached/full readouts. All 118 metric rows, 332 candidate positions and three old fixed references independently match; maximum metric error 1.22e-15.

Private pixel review covers original base/joint inputs, corrected order/joint controls, all final updates and V03 references. Private RGB, Query text, GT coordinates, weights and optimizer state are excluded from public output. Anonymous per-case results preserve all positive, negative, neutral and unknown values.

Actual teacher cost: 107,320 input tokens, 65,536 visual tokens, 3,245 output tokens; 76.674 s summed request time. Includes repairs; excludes cold load, asset verification and preparation. This is not end-to-end deployment latency.

Finite diagnosis complete. Stable bidomain spatial benefit is not established. No automatic expansion, new temporal training, online carry, hyperparameter search or CURRENT promotion. QA-interval anchor sampling remains a separate unexecuted ablation.